Literature Review

E-Commerce Data Engine & AI Assistant with Integrated Data Quality Monitoring

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Scope note: This review summarises the concepts and tools already used in the project proposal. Tool-specific decisions remain provisional until confirmed in System Analysis & Design.

1. Introduction

E-commerce platforms generate structured data (orders, order items, payments, products, sellers, delivery timestamps) and unstructured or semi-structured data (customer review comments). Turning these into dependable insights requires repeatable pipelines, analytical modelling, and controls that communicate when data is not trustworthy.

This review covers ETL/ELT, dimensional modelling, processing and orchestration, storage, BI, Portuguese review sentiment, RAG, and data-quality monitoring. It then identifies the project's intended contribution: connecting quality findings to the reliability status of downstream analytics and AI.

2. Data pipelines and warehousing

2.1 ETL and ELT: ETL transforms data before loading it into the analytical target; ELT loads data first and transforms within the target. Either pattern must address missing values, duplicates, inconsistent formats, and schema changes. For this project, use staged ingestion, explicit transformations, and validated warehouse loading so failures are visible and repeatable.

2.2 Dimensional modelling: Kimball and Ross [1] describe dimensional modelling with fact tables for measurable business events and dimensions for descriptive context. A proposed star schema may include order/item sales facts and payment/shipping facts, with Date, Product, Customer, Seller, and Category dimensions. Define each fact table's grain before creating joins; avoid joining multiple one-to-many tables in a way that double-counts sales or payments.

2.3 Processing: Pandas and Spark: Pandas is suitable for moderate data volumes that fit comfortably in memory. Apache Spark [2] distributes processing and is useful when scale or workload justifies cluster-style execution. Since Olist is a bounded public dataset, Pandas is the default for a simpler, reproducible implementation; Spark remains an optional scalability path rather than a mandatory duplicate implementation.

2.4 Orchestration: Apache Airflow represents workflows as DAGs and supports scheduling, dependencies, retries, logs, and task monitoring. DQ checks should be explicit pipeline tasks, and downstream warehouse publication should depend on the result of required critical checks.

2.5 Storage: PostgreSQL supports relational, typed data and SQL analytics. MongoDB can store flexible review documents, although keeping reviews in PostgreSQL is also viable. If both stores are used, document which system is authoritative for each entity and how IDs and refreshes are reconciled. Azure is the intended cloud target, but the exact

service must be selected based on access, cost, and team capability.

3. Analytics and AI

3.1 Business intelligence: A dashboard converts warehouse records into business metrics. Streamlit is a proposed Python-based dashboard framework; its selection should be confirmed and recorded. Define top-selling products in two separate ways: units sold and item revenue. Define average price as item price, excluding freight unless the metric is explicitly called total order cost.

3.2 Sentiment analysis: Sentiment analysis classifies text into categories such as positive, neutral, and negative. Olist reviews are in Portuguese, so a Portuguese-capable model such as BERTimbau [12] or a validated multilingual model is needed. Evaluate performance on a manually labelled sample, report the sample size and class-level precision/recall or F1, and do not assume an English-only model is suitable. Reviews without comment text cannot receive text-model sentiment; a review-score mapping (4–5 positive, 3 neutral, 1–2 negative) can be shown separately as a score-based proxy.

3.3 Retrieval-Augmented Generation: Lewis et al. [5] introduced RAG, combining a language model with retrieved external information. In this project, retrieval should be split by task: SQL queries for exact structured conditions and aggregations (budget limits, category, units sold, revenue, review scores, synthetic stock); semantic retrieval for review text or explanatory documents. Vector similarity alone is not reliable for numeric filters such as “under 200 BRL.” The assistant must use retrieved evidence, distinguish unavailable attributes, and abstain when the dataset cannot answer.

3.4 Limitations and evaluation: A RAG response can still be wrong if retrieval, source data, or interpretation is wrong. Evaluate with a fixed question set that includes answerable questions, no-match questions, unsupported requests, and unreliable-data scenarios. Measure groundedness/faithfulness, retrieval relevance, correct abstention, and response latency. A Trust/DQ status should be available to the assistant and should prevent unsupported confidence when critical data checks fail.

4. Data quality and monitoring

4.1 Dimensions: Wang and Strong [6] describe data quality as multi-dimensional, including accuracy, completeness, consistency, and timeliness. This project focuses on schema integrity, statistical plausibility, cross-system consistency, and the quality state of downstream data.

4.2 Automated validation: Schelter et al. [7] and Breck et al. [8] motivate automated validation as part of data/ML pipelines rather than a manual afterthought. This project should validate raw/staging and curated layers, attach rule IDs and severity, and preserve test results for auditability.

4.3 Anomalies and drift: Chandola et al. [9] survey anomaly-detection methods. Z-scores are explainable for roughly well-behaved numeric distributions but can be sensitive to skew and outliers; rolling baselines must be grouped appropriately (for example, category or date window). Isolation Forest [10] can be an optional multivariate detector. Gama et al. [11] discuss concept drift, motivating monitoring for changes in data distributions over time.

4.4 Tool comparison: Great Expectations, Deequ, TFX Data Validation, Soda, and dbt tests provide useful expectations, constraints, or schema/drift checks. The project should not claim that these tools cannot perform cross-system checks in general; instead, the project-specific contribution is to combine its chosen checks with explicit downstream trust status for both BI and the assistant.

5. Research gap and project contribution

Existing validation frameworks provide building blocks for dataset and pipeline checks. This project focuses on an integrated demonstration where (1) schema, statistical, and cross-pipeline checks run as part of a scheduled workflow; (2) findings are classified as Warning or Critical with defined batch behavior; and (3) a Trust/Health view communicates whether data consumed by BI and the assistant is Healthy, Warning, or Critical. The contribution is the integration and policy layer, not a claim that no existing tool can do cross-table validation.

6. Conclusion

The reviewed work supports dimensional modelling, scheduled orchestration, language-model retrieval, sentiment classification, and automated data validation. For this project, reliability depends on applying each technique appropriately: exact structured filters should use SQL, sentiment models must support Portuguese, thresholds must be calibrated against observed data, and downstream consumers must be informed when data quality is degraded.

7. References

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Reference and data notes

Verify access dates and bibliographic details against the source pages before formal academic submission. The original draft cites some references by number without a complete bibliography; the list above supplies the intended sources based on the cited authors/topics. Dataset counts and missing-text percentages should be recomputed from the exact downloaded dataset version and reported as project observations, not universal constants.